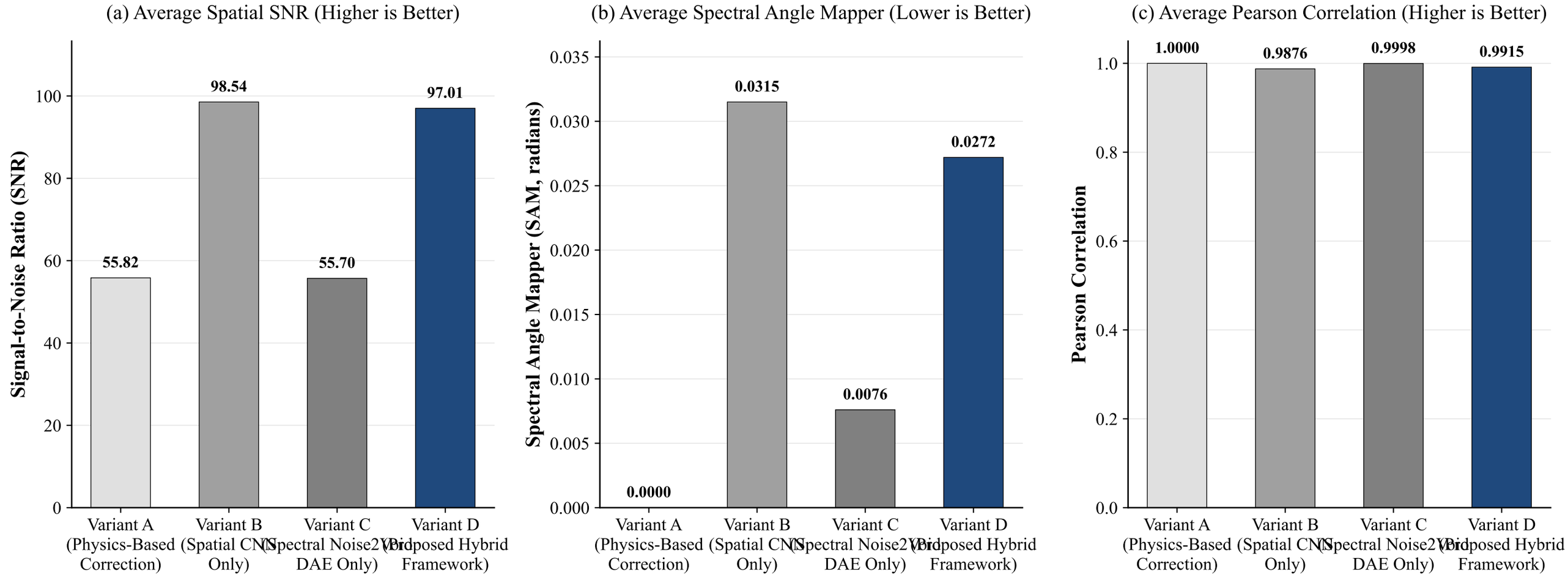
Ablation Analysis of the Proposed Spatial–Spectral Hybrid Denoising Framework



Key Findings

- Variant A (Physics-Based Correction) serves as the reference baseline and provides initial removal of deterministic sensor artifacts.
- Variant B (Spatial CNN) achieves the highest spatial enhancement by effectively suppressing residual stripe noise and improving SNR.
- Variant C (Spectral Noise2Void DAE) preserves spectral characteristics, producing lower SAM and higher Pearson correlation than the spatial module alone.
- Variant D (Proposed Hybrid Framework) combines the strengths of both learning modules, achieving the best balance between spatial restoration and spectral fidelity.

** Variant A is the reference stage; therefore, SAM = 0 and Pearson correlation = 1 when the physics-corrected cube is compared with itself.*